

Finite Fields

Question 1

What is the smallest n for which $\mathbb{F}_{3^n}$ contains a non-trivial 5-th root of unity?

Solution. Since 5 is a prime, any non-trivial 5-th root of unity is a primitive 5-th root of unity.

In order that $\mathbb{F}_{3^n}$ contains a primitive 5-th root of unity, it is necessary and sufficient that 5 divides $3^n - 1$.

Thus, n is the order of 3 modulo 5.

We calculate the powers of $(3 \bmod 5)$ as:

$$\begin{array}{ll} (3 \bmod 5)^1 = 3 & (3 \bmod 5)^2 = 4 \\ (3 \bmod 5)^3 = 2 & (3 \bmod 5)^4 = 1 \end{array}$$

Thus we see that $n = 4$.

Question 2

What is the smallest n for which the polynomial $X^3 + X + 1$ has all its roots in the field $\mathbb{F}_{2^n}$?

Solution. We note that $X^3 + X + 1$ is irreducible (seen earlier) and of degree 3. Thus n is at least 3.

Elements of $\mathbb{F}_{2^3}$ satisfy the polynomial $X^8 + X$. Thus, we write the long division of $X^7 + 1$ by $P(X) = X^3 + X + 1$ over $\mathbb{F}_2$:

$$\begin{aligned} X^7 + 1 &= X^4 \cdot (X^3 + X + 1) + X^5 + X^4 + 1 \\ &= X^4 P(X) + X^2 \cdot (X^3 + X + 1) + X^4 + X^3 + X^2 + 1 \\ &= (X^4 + X^2) P(X) + X \cdot (X^3 + X + 1) + X^3 + X + 1 \\ &= (X^4 + X^2 + X + 1) P(X) \end{aligned}$$

This shows that all the roots of this polynomial lie in $\mathbb{F}_{2^3}$.

We can also argue as follows. The field $\mathbb{F}_{2^3}$ is a *Galois* cyclic extension of $\mathbb{F}_2$ since it is the splitting field of $X^8 + X$ over $\mathbb{F}_2$. It follows that if one root of $X^3 + X + 1$ belongs to the field $\mathbb{F}_{2^3}$, then *all* the roots of this polynomial lie in this field.